

CSCI 152 Performance and Data Structures

Lab 3

Deadline: June 28, 2025 at 23:59

This lab is aimed to prepare you for reference-based data structures. Some samples and parts of the code may be recognized from the video lectures. The task is to implement a Node class based on the provided Node.h (do not change in Part 1 and Part 2). There are three packages for three parts of the lab. Each package contains Node.h, Main.cpp, and a makefile. You are encouraged to test with more test cases in Main.cpp to check the robustness of the class. The result of the method printNode() in all parts is commented in Main.cpp, so you may use custom messages accordingly.

Part 1: Singly Node

You need to prepare a Node.cpp file with the provided Node.h for a singly Node. It contains only an integer value and a pointer link to the next node. You need to submit the Part-1-Node.cpp file.

Part 2: Doubly Node

You need to prepare a Node.cpp file with the provided Node.h for a doubly Node. It contains an integer value and links to the previous and next nodes. Pay attention to the constructor: What is the value of the previous node if it is not passed? What if only the value is passed to the constructor? As this Node class has next and previous links, it may be used to implement doubly linked lists and deques. You need to submit the Part-2-Node.cpp file. When you set node A as previous to node B, you also need to set the next pointer of node A to node B. Similarly, when you set node B as the next node of node A, you need to set node A as the previous node of node B.

Part 3: Doubly Node for Different Data Types

You need to generalize the doubly node to support values of different data types. In Part 3, you need to create your own Node.h, which contains both the class declaration and implementation. Main.cpp and a makefile are provided. You need to submit the Part-3-Node.h file. When you set node A as previous to node B, you also need to set the next pointer of node A to node B. Similarly, when you set node B as the next node of node A, you need to set node A as the previous node of node B.